Akoizo Original Practice - 2027 Rotation Form 3

Division C | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Distinguish an active lidar measurement from a passive optical radiometer measurement.

Answer: _____
_____
_____

2. (2 points) Explain how spectral resolution differs from radiometric resolution.

Answer: _____
_____
_____

3. (2 points) Apply Wien's law with b=2.90e-3 m K to a 290 K blackbody. Find peak wavelength.

Answer: _____
_____
_____

4. (2 points) Why is space-based gravimetry relevant to changing ice mass?

Answer: _____
_____
_____

5. (2 points) Explain why a geostationary observer and a polar-orbiting observer provide different sampling patterns.

Answer: _____
_____
_____

6. (2 points) Compute net incoming minus outgoing top-of-atmosphere energy flux.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 345 W/m^2, reflected solar flux 103 W/m^2 and outgoing longwave flux 236 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

Answer: _____
_____
_____

7. (2 points) A JASON teaching series shows 12, 15 and 18 mm at yearly intervals. Find its endpoint slope.

Answer: _____
_____
_____

8. (2 points) A GRACE teaching estimate changes from 0 to -60 gigatonnes over 2 years. Find mean mass-change rate.

Answer: _____
_____
_____

9. (2 points) Why can the JASON and GRACE observations not be substituted for one another?

Answer: _____
_____
_____

10. (2 points) An OCO-2 teaching column-CO2 series changes from 403 to 408 ppm. Find the increase.

Answer: _____
_____
_____

11. (2 points) A MODIS scene is masked as cloud-covered. Explain why treating missing pixels as zero aerosol concentration is wrong.

Answer: _____
_____
_____

12. (2 points) A CrIS teaching dataset supplies temperature estimates while an instrument map supplies radiative flux. Explain why temperature and energy flux have different units and meanings.

Answer: _____
_____
_____

13. (2 points) Explain a climate pathway from increasing atmospheric CO2 to changing outgoing longwave radiation, with surface temperature initially fixed.

Answer: _____
_____
_____

14. (2 points) A supplied ENSO index changes sign. Why should regional rainfall impacts be evaluated from regional data rather than assumed identical everywhere?

Answer: _____
_____
_____

15. (2 points) Distinguish a carbon flux from a carbon reservoir in a climate budget.

Answer: _____
_____
_____

16. (2 points) Why can changing vegetation distributions affect both carbon cycling and surface energy exchange?

Answer: _____
_____
_____

17. (2 points) Calculate reflected/incoming solar flux as a dimensionless albedo.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 345 W/m^2, reflected solar flux 103 W/m^2 and outgoing longwave flux 236 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

Answer: _____
_____
_____

18. (2 points) For fixed emissivity, a blackbody temperature doubles. By what factor does emitted power per unit area change?

Answer: _____
_____
_____

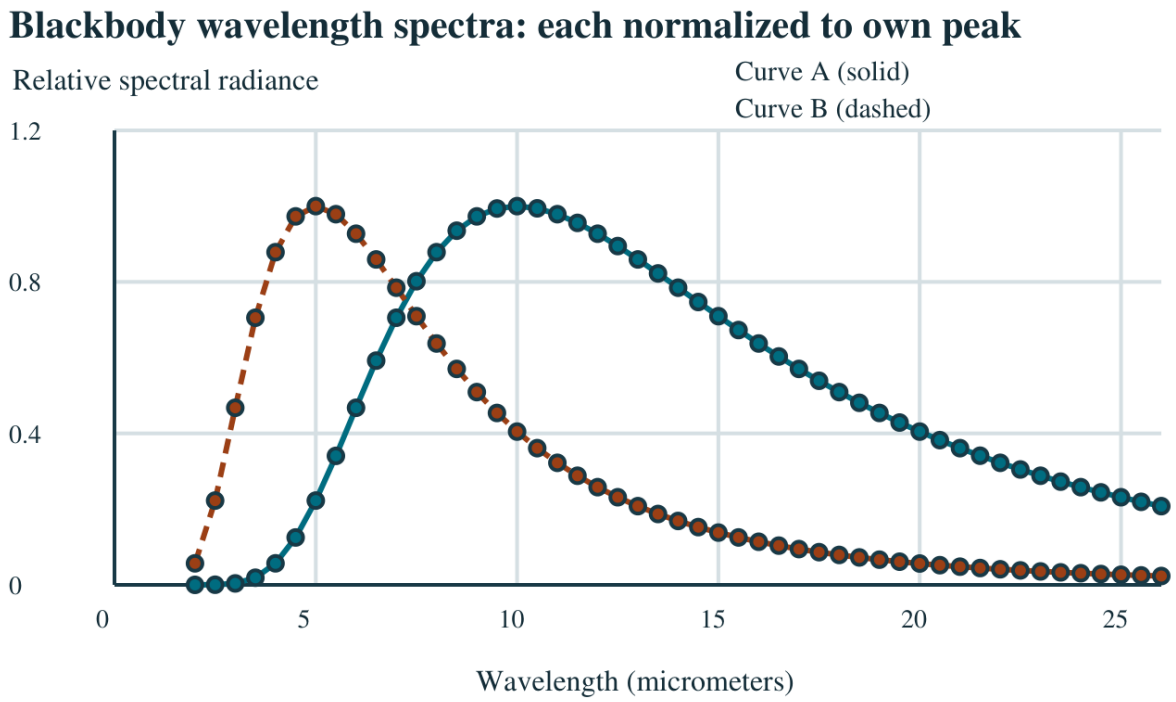
19. (2 points) A supplied linear energy anomaly grows from 3 to 12 W/m^2 over 3 years. Find its slope.

Answer: _____
_____
_____

20. (2 points) Explain the ice-albedo feedback after an initial warming perturbation.

Answer: _____
_____
_____

21. (4 points) Use Figure F1. Use the peak of A and b=2900 micrometer K to estimate its temperature.



F1. Normalized thermal spectra (original synthetic teaching figure)

Answer: _____
_____
_____

22. (3 points) Use Figure F1. Which curve represents a higher-temperature blackbody? Explain.

Refer to F1. Normalized thermal spectra (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

23. (3 points) Use Figure F1. Why cannot equal plotted peak heights establish equal absolute radiance?

Refer to F1. Normalized thermal spectra (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (5 points) Use Figure F2. Calculate the endpoint JASON sea-level trend.

Synthetic JASON and GRACE observations

Year	JASON sea level (mm)	GRACE ice-mass change (Gt)
0	0	0
1	5	-30
2	10	-60
3	15	-90

F2. Satellite trend panels (original synthetic teaching figure)

Answer: _____
_____
_____

25. (5 points) Use Figure F2. Calculate the endpoint GRACE ice-mass trend.

Refer to F2. Satellite trend panels (original synthetic teaching figure) shown with the first question in this figure group.

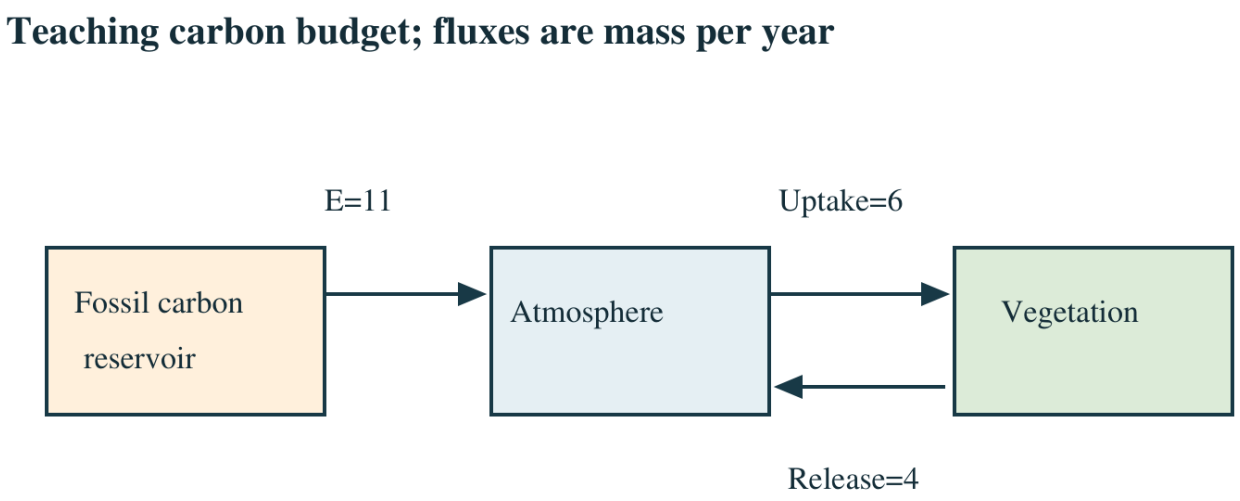
Answer: _____
_____
_____

26. (4 points) Use Figure F2. Why cannot a sea-level trend in mm/year simply replace the GRACE trend?

Refer to F2. Satellite trend panels (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (3 points) Use Figure F3. Find the net annual vegetation carbon gain from the shown arrows.



F3. Carbon reservoirs and transfers (original synthetic teaching figure)

Answer: _____
_____
_____

28. (3 points) Use Figure F3. Find the net atmospheric gain in this simplified budget.

Refer to F3. Carbon reservoirs and transfers (original synthetic teaching figure) shown with the first question in this figure group.

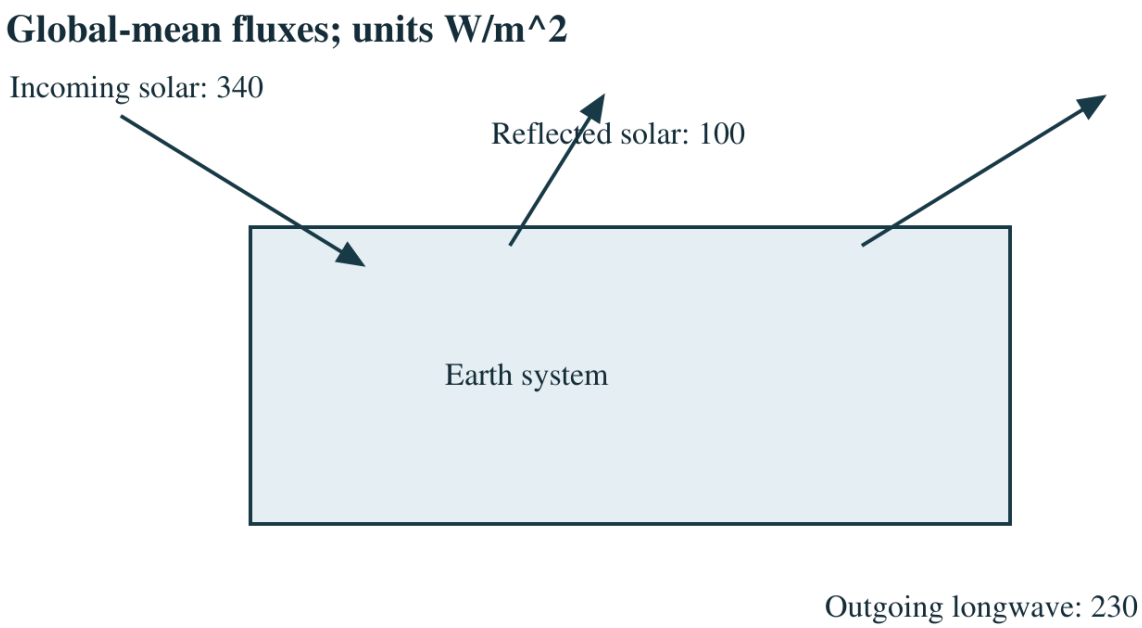
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Distinguish a box quantity from an arrow quantity in this diagram.

Refer to F3. Carbon reservoirs and transfers (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (3 points) Use Figure F4. Calculate the net incoming energy flux.



F4. Top-of-atmosphere energy budget (original synthetic teaching figure)

Answer: _____
_____
_____

31. (3 points) Use Figure F4. Calculate the implied planetary albedo.

Refer to F4. Top-of-atmosphere energy budget (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. If outgoing longwave and incoming solar initially remain fixed while reflected solar decreases, describe the immediate change in net heating.

Refer to F4. Top-of-atmosphere energy budget (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____